Multimedia Signal Processing - 2nd Module

June, 29 2021

1. **Question (8 pts)** Consider a discrete source X that emits as symbols the musical notes from the alphabet $[C, D, E, F, G, A, B]$. Consider the problem of encoding in a lossless fashion the following sequence of notes:
$$x = \{C, C, G, G, A, A, G, G, F, F, E, E, D, D, C, C, C, C\}.$$
   (a) Estimate the pmf of the source by considering the sequence x as a representative example.
   (b) Compute the entropy $H(X)$ of the source X.
   (c) Derive the Huffman code for the sequence x, showing a picture of the Huffman tree.
   (d) Compute the average code length of the Huffman code.
   (e) Encode the sequence x with the derived Huffman code (i.e., write the extension C^*).
   (f) Consider the case of encoding the same sequence by taking groups of two symbols at a time. Compute the theoretical minimum amount of bit per symbol needed to losslessy encode the signal.
   (g) Repeat the last point in the case you consider three symbols at a time, comparing the results with the previously obtained ones.

2. **IF ONLINE: Question (7 pts)**
   Oral examination.

3. **IF IN CLASS: Question (5 pts)** Consider the problem of audio perceptual coding.
   - Describe the concept of absolute threshold of hearing, and sketch it clearly explaining what's represented on the axes.
   - Describe how it is possible to obtain a modified threshold of hearing for coding using MPEG 1 psychoacoustic model 1.

4. **IF IN CLASS: Question (2 pts)** Clearly explain the concept of motion estimation and motion compensation in video sequences (hint: help yourself with some sketches).

5. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

- [T] [F] Consider a source X with alphabet $A = \{a, b, z\}$. Consider the following associated code: $C(a) = 0$, $C(b) = 01$, $C(z) = 10$. This code may have been obtained using Huffman coding scheme.
- [T] [F] Consider a discrete memoryless source X with alphabet $A = \{a, b, p, k\}$. The entropy of X can be 3.
- [T] [F] Consider a discrete memoryless source X with variance σ_x^2. According to rate-distortion theory, it is possible to encode X with distortion greater than σ_x^2 at any rate.
- [T] [F] Consider a close-loop DPCM coding scheme. It is possible to use an open-loop approximation to estimate which prediction rule should be used.
- [T] [F] In a transform coding scheme, the decoder must know which transform was used by the encoder.
- [T] [F] Consider the problem of speech coding. If we us a DPCM, we always need both a long-term and short-term predictor.
- [T] [F] Consider masking effects in audio coding. Tones can mask noise, but not the vice versa.
- [T] [F] Video coding schemes always use motion estimation and compensation.

6. **Question (Matlab - 10pts)**

Load the default Matlab image `peppers.png` using the command `imread('peppers.png')`.

(a) Display the image.

(b) Extract the blue chrominance component of the image knowing that the RGB-to-YCbCr color space transformation is $\begin{bmatrix} x_{\mathrm{y}} \\ x_{\mathrm{Cb}} \\ x_{\mathrm{Cr}} \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.5 \\ 0.5 & -0.419 & -0.0813 \end{bmatrix} \begin{bmatrix} x_{\mathrm{R}} \\ x_{\mathrm{G}} \\ x_{\mathrm{B}} \end{bmatrix}$.

(c) Quantize the blue chrominance with a uniform scalar quantizer that uses B bit per symbol. Consider all following B values in $\{2, 3, 4, 5, 6, 7, 8\}$.

(d) For each obtained quantization step, compute the PSNR between the quantized chrominance and the original one.

(e) For each obtained quantization step, compute the entropy of the quantized chrominance.

(f) Plot the Rate distortion curve (use entropy as rate and PSNR as distortion).

(g) Plot the Rate distortion curve (use B as rate and PSNR as distortion).